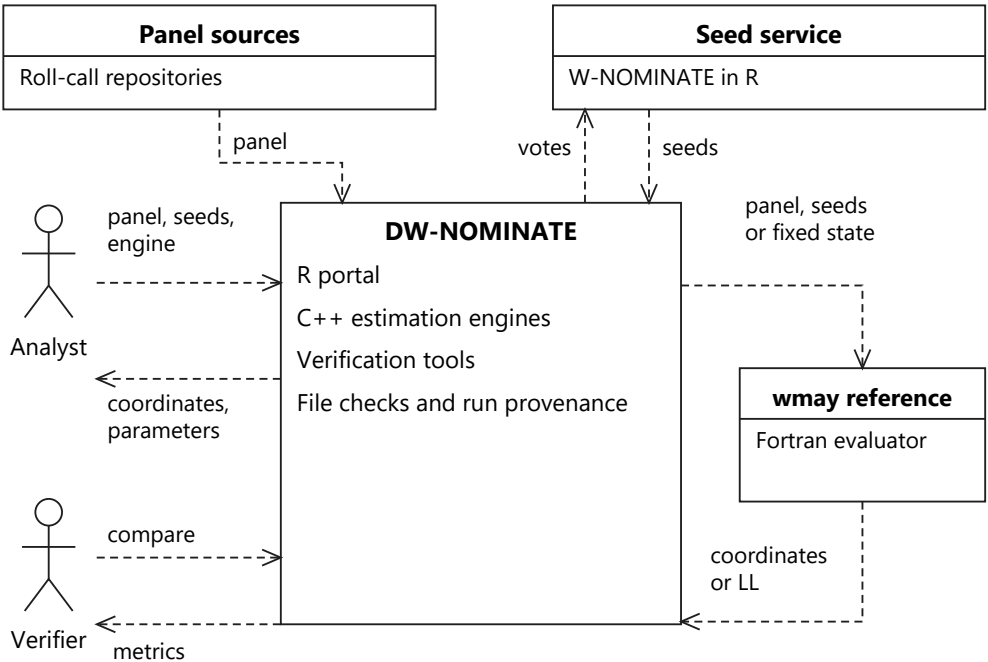
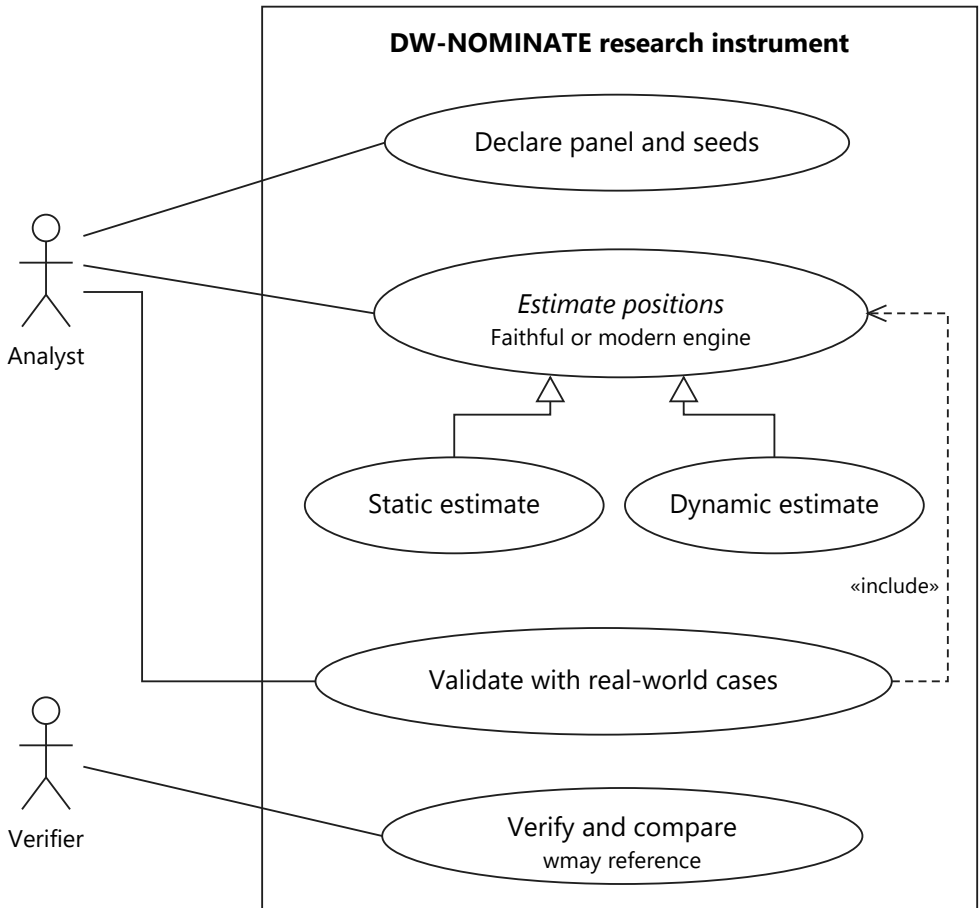
Instrument context

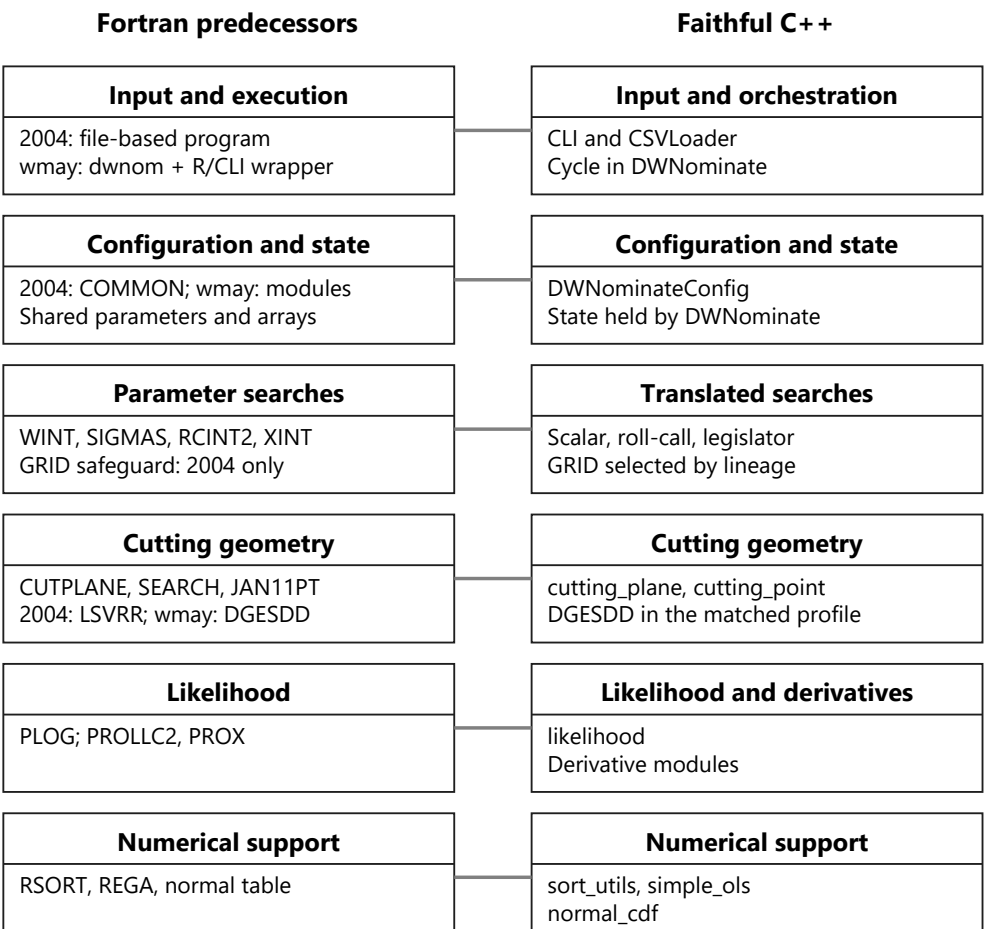


Dashed arrows: data flow. LL: log-likelihood.

Research use cases

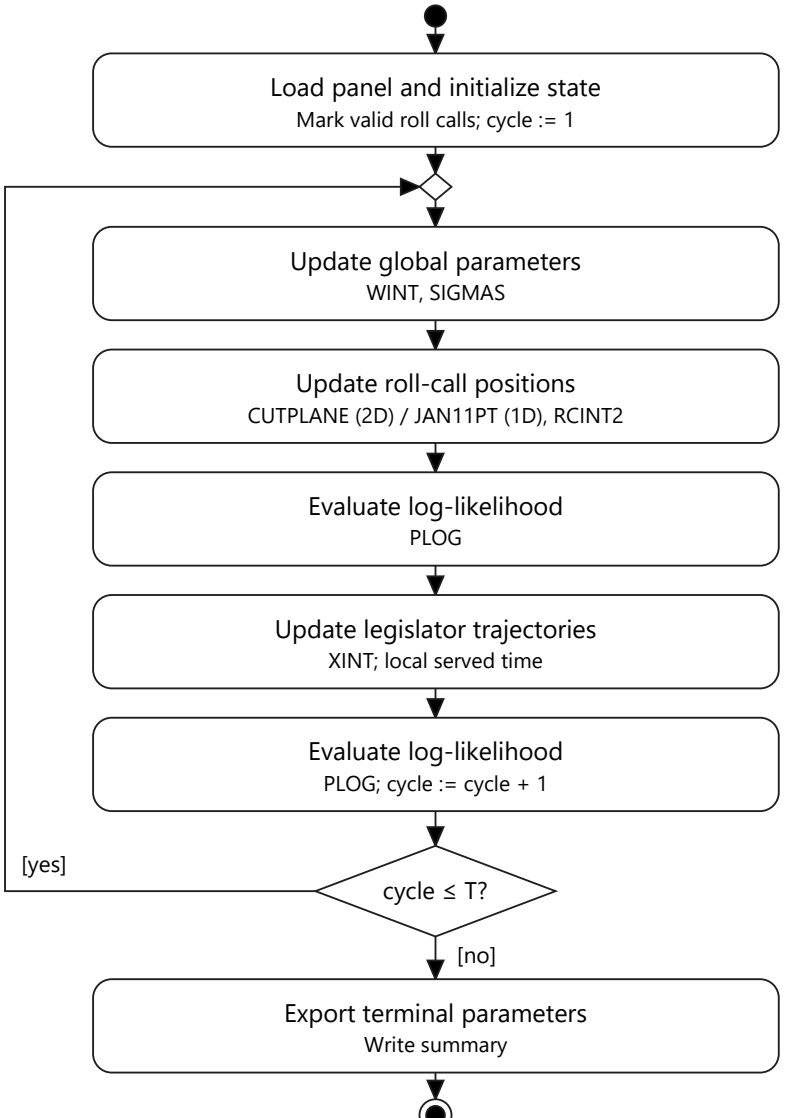


Legacy responsibility mapping



Each horizontal link maps a responsibility, not a dependency.

Fixed-cycle estimation



Scope: faithful in wmay mode; 2D, T = 4. No convergence test.

Known limitation: faithful summary mixes terminal LL and earlier classification.

Optimization mechanisms

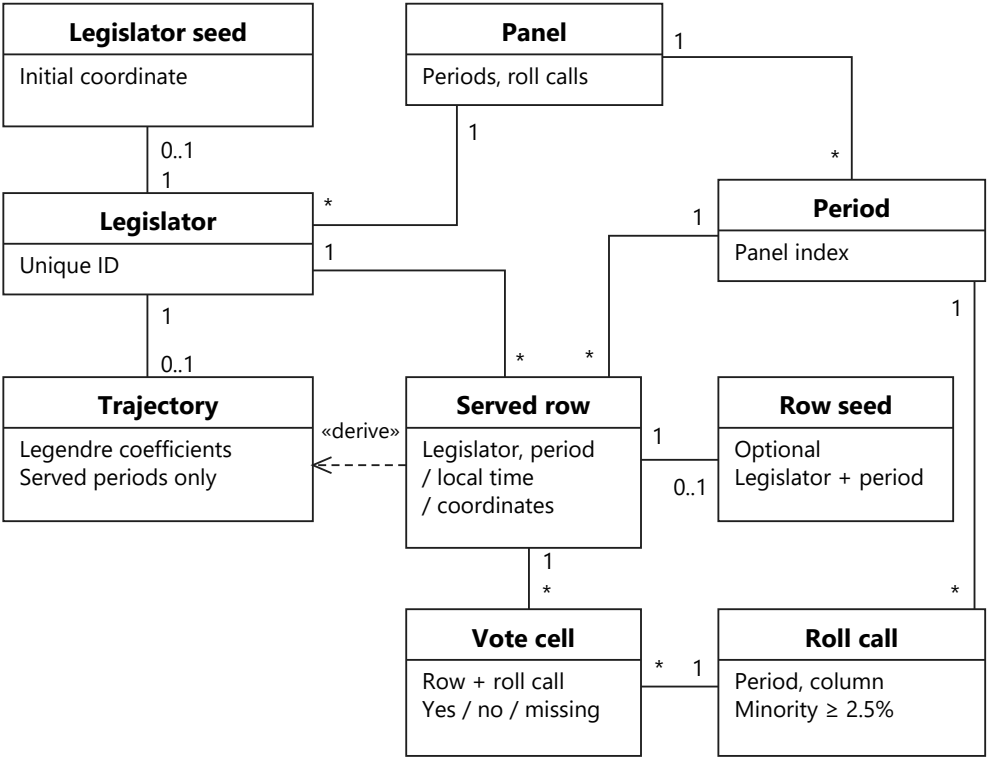
Controlled profile: local scalar searches, strict arithmetic, COBYLA blocks.

Parameter	Faithful	Modern	Constraint
2nd-dimension weight	WINT grid	BOBYQA scalar	Local WINT interval; 2D only
Beta	SIGMAS grid	BOBYQA scalar	Local SIGMAS interval
Roll call	RCINT2 term-wise search	COBYLA block	Midpoint norm ≤ 1
Legislator	XINT term-wise search	COBYLA block	Constant-term norm ≤ 1

Global scalar search, SLSQP and hybrid blocks are outside this profile.

Replication removes global scalar bounds; the local search interval remains.

Data and served time

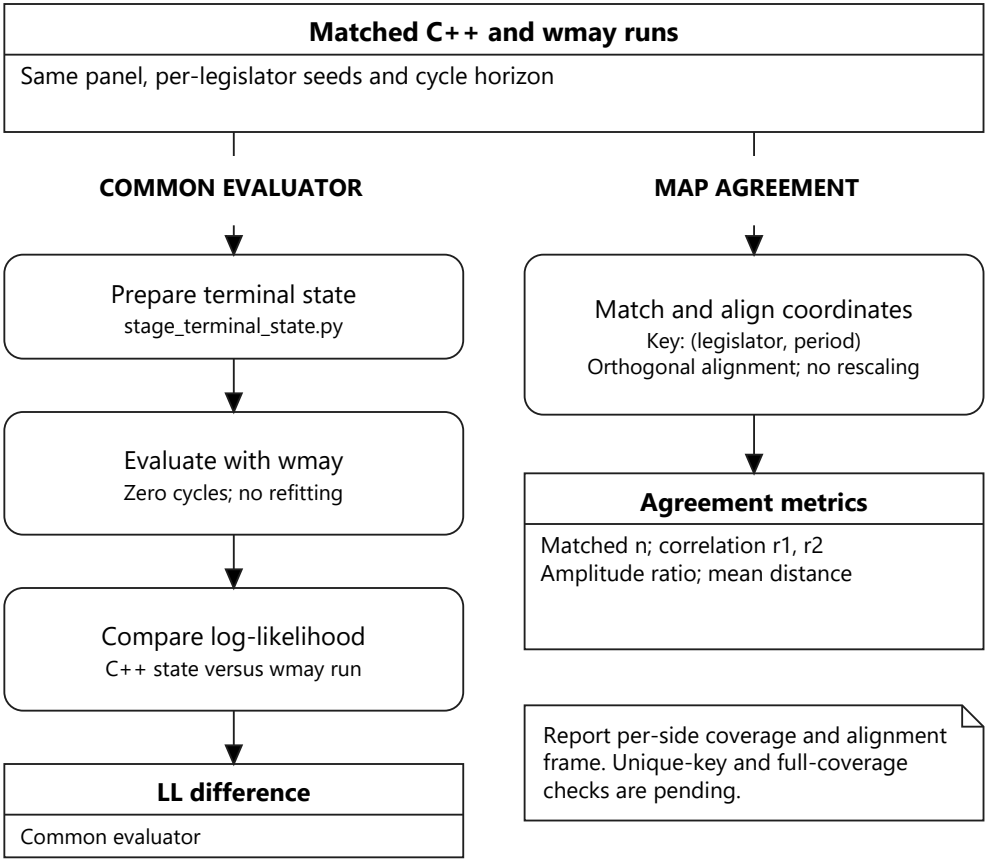


Example: one legislator serves periods 2 and 4

Period 1	**Period 2**	Period 3	**Period 4**	Period 5
no row	t = -1	no row	t = +1	no row

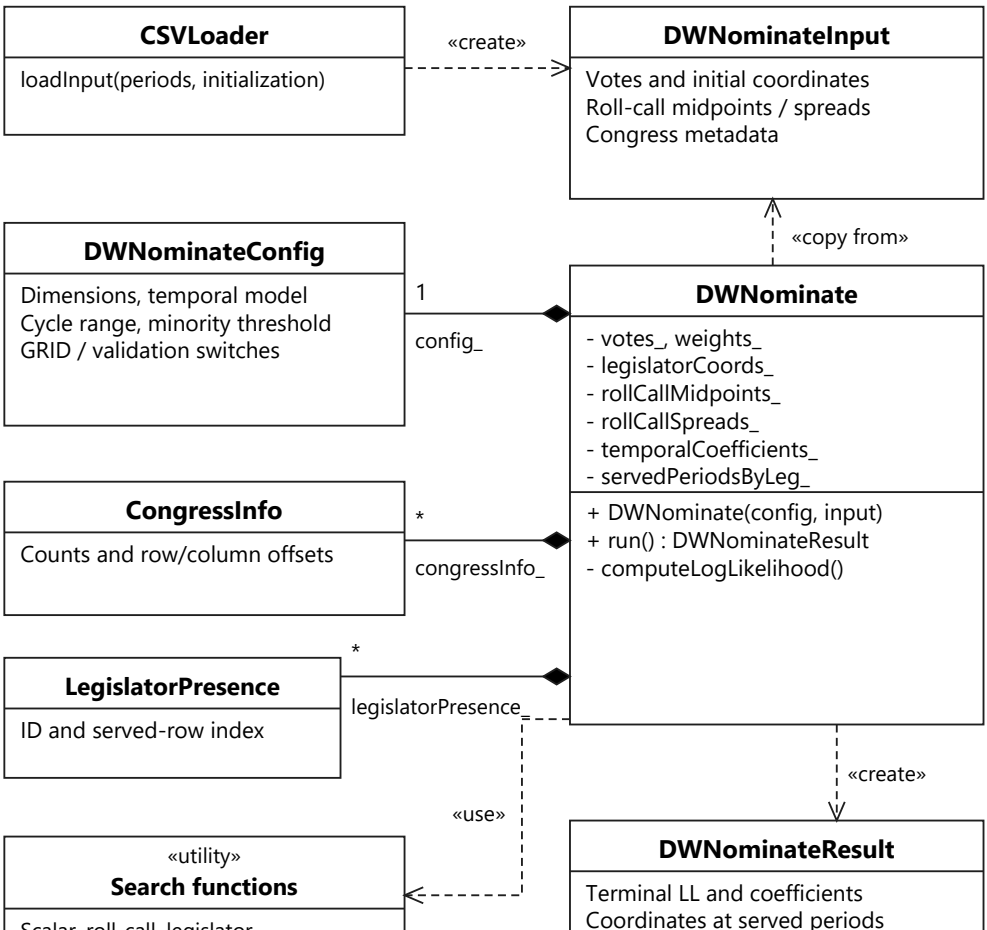
Two served periods → constant trajectory. Linear / quadratic / cubic need 5 / 6 / 7.

Two complementary checks



Reference here: wmay. This protocol does not establish original-2004 equivalence.

Faithful engine: state ownership



Filled diamond: ownership by value. Selected members; input is not retained.

One estimation request

